

Dildar Ali

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About Me

I work on the design and theoretical analysis of approximation algorithms for NP-hard combinatorial optimization problems, primarily through the lens of submodular and bisubmodular function maximization, with applications to influence maximization, regret minimization, and algorithmic data management. I completed my Ph.D. in Computer Science and Engineering at the Indian Institute of Technology Jammu, advised by [Prof. Suman Banerjee](#) and [Prof. Yamuna Prasad](#). I am now seeking a postdoctoral position to extend this line of work toward scalable, provably-guaranteed algorithms for large-scale combinatorial and data-driven decision problems.

Research Interests

AI for Combinatorial Optimization · Stochastic & Robust Optimization · Approximation Algorithms · Algorithmic Data Management · Reinforcement Learning · Multi-Agent Decision Making

Education

Ph.D. in Computer Science and Engineering *Indian Institute of Technology (IIT) Jammu* Jan 2022 – Apr 2026

- Thesis: *Influential Slot Selection and Regret Minimization in Billboard Advertisement*
- Supervisor: Prof. Suman Banerjee · Co-Supervisor: Prof. Yamuna Prasad, IIT Jammu, India

M.Tech in Computer Science and Engineering *Aliah University, India* Jul 2018 – Jun 2020

- CGPA: 9.16/10 (University 2nd Rank Holder)
- Thesis: *A Deep Learning Approach to Improve Effectiveness and Efficiency for Time Series Prediction*

Publications

21 peer-reviewed publications: 5 journal articles, 16 conference papers, including AAAI, PAKDD ($\times 2$), DEXA ($\times 2$), ADBIS, WISE, SEA, ADC ($\times 2$), ACM SAC, ICAART, EUMAS, ADMA, and IKDD CODS.

Journal Articles

- [J6] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Joint Slot and Tag Selection in Billboard Advertising: A Bi-Submodular Maximization Approach.” [Submitted, **Knowledge and Information Systems**]
- [J5] Dildar Ali, Suman Banerjee, Rajibul Islam. “Trajectory-Driven Multi-Product Influence Maximization in Billboard Advertising.” [Accepted, **Journal of Combinatorial Optimization** [Link](#)]
- [J4] Poonam Sharma, Dildar Ali, Suman Banerjee. “A Regret-Aware Framework for Effective Social Media Advertising.” **International Journal of Data Science and Analytics**. [Link](#)
- [J3] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Minimizing Regret in Billboard Advertisement under Zonal Influence Constraint.” **Knowledge and Information Systems (KAIS)**. [Link](#)
- [J2] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Influential Billboard Slot Selection using Spatial Clustering and Pruned Submodularity Graph.” **International Journal of Data Science and Engineering**, 1–22. [Link](#)
- [J1] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Toward Regret-Free Slot Allocation in Billboard Advertisement.” **International Journal of Data Science and Analytics** (2024): 1–25. [Link](#)

Conference Papers

- [C18] Dildar Ali, Paras Dhatwalia, Suman Banerjee. “Two-Sided Online Bipartite Matching for Billboard Ad Allocation: A Deep Reinforcement Learning Approach.” [Submitted, **ACML 2026**]
- [C17] Dildar Ali, Abishek Salaria, Ansh Jasrotia, Suman Banerjee. “Tag-specific Regret Minimization Problem in Outdoor Advertising.” [Submitted, **IKDD CODS 2026**] [Link](#)
- [C16] Dildar Ali, Ansh Jasrotia, Abishek Salaria, Suman Banerjee. “Approximation Algorithms for Budget Splitting in Multi-Channel Influence Maximization.” The 24th **International Symposium on Experimental Algorithms (SEA 2026)**. [Link](#)

- [C15] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Approximately Bisubmodular Regret Minimization in Billboard and Social Media Advertising.” The 30th **Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)**. [Link](#) 
- [C14] Aditi Anand, Dildar Ali, Suman Banerjee. “Fairness Driven Multi-Agent Path Finding Problem.” The 18th **International Conference on Agents and Artificial Intelligence (ICAART 2026)**. [Link](#) 
- [C13] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Balanced Popularity in Multi-Product Billboard Advertisement.” The 36th **Australasian Database Conference (ADC 2025)**. [Link](#) 
- [C12] Dildar Ali, Rajibul Islam, Suman Banerjee. “Multi-product Influence Maximization in Billboard Advertisement.” The 13th **ACM IKDD International Conference on Data Science (CODS 2025)**. [Link](#) 
- [C11] Aditi Anand, Suman Banerjee, Dildar Ali. “Scalable Submodular Policy Optimization via Pruned Submodularity Graph.” The 22nd **European Conference on Multi-Agent Systems (EUMAS 2025)**. [Link](#) 
- [C10] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Group Trip Planning Query Problem with Multimodal Journey.” The 36th **International Conference on Database and Expert Systems Applications (DEXA 2025)**. [Link](#) 
- [C9] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Influential Slot and Tag Selection in Billboard Advertisement.” The 36th **International Conference on Database and Expert Systems Applications (DEXA 2025)**. [Link](#) 
- [C8] P. Sharma, Dildar Ali, Suman Banerjee. “Minimizing Regret in Social Media Advertisement.” The 12th **IEEE International Conference on Big Data (BigData 2024)**, pp. 473–478. [Link](#) 
- [C7] Dildar Ali, Suman Banerjee, Shweta Jain, Yamuna Prasad. “Fairness Driven Slot Allocation Problem in Billboard Advertisement.” The 29th **Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2025)**, LNCS vol. 15871. [Link](#) 
- [C6] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Multi-Slot Tag Assignment Problem in Billboard Advertisement.” The 35th **Australasian Database Conference (ADC 2024)**, pp. 3–16. [Link](#) 
- [C5] Dildar Ali, Harishchandra Kumar, Suman Banerjee, Yamuna Prasad. “An Effective Tag Assignment Approach for Billboard Advertisement.” The 25th **International Conference on Web Information Systems Engineering (WISE 2024)**, pp. 159–173. [Link](#) 
- [C4] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Influential Billboard Slot Selection Under Zonal Influence Constraint.” The 28th **European Conference on Advances in Databases and Information Systems (ADBIS 2024)**, pp. 93–106. [Link](#) 
- [C3] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Regret Minimization in Billboard Advertisement under Zonal Influence Constraint.” The 39th **ACM/SIGAPP Symposium on Applied Computing (ACM SAC 2024)**, pp. 329–336. [Link](#) 
- [C2] Dildar Ali, A.K. Bhagat, Suman Banerjee, Yamuna Prasad. “Efficient Algorithms for Regret Minimization in Billboard Advertisement” (Student Abstract). The 37th **AAAI Conference on Artificial Intelligence (AAAI 2023)**, 37(13), 16148–16149. [Link](#) 
- [C1] Dildar Ali, Suman Banerjee, Yamuna Prasad. “Influential Billboard Slot Selection Using Pruned Submodularity Graph.” The 18th **International Conference on Advanced Data Mining and Applications (ADMA 2022)**. [Link](#) 

Research Contributions

- Developed approximation algorithms for NP-hard combinatorial optimization problems based on submodular and bisubmodular optimization with provable guarantees.
- Designed scalable algorithms for influence maximization and regret minimization under spatial, budget, and fairness constraints.
- Proposed efficient optimization techniques, including Pruned Submodularity Graphs, to accelerate large-scale submodular optimization.
- Extended optimization frameworks to multi-product, multi-channel, and multi-agent decision-making problems with theoretical analysis.
- Authored 21 peer-reviewed publications (5 journal articles, 16 conference papers) at venues including AAAI, PAKDD, DEXA, SEA, ADBIS, WISE, ACM SAC, ICAART, and EUMAS.

Experience

Post Doctoral Fellow *National Tsing Hua University (NTHU), Taiwan*

*Hsinchu, Taiwan
March 2026 – July 2026*

- Project: “Advanced Information Technology Research on Multimodal Language Reinforcement Models”, (NSTC 115-2811-E-007-006), funded by National Science and Technology Council (NSTC), Taiwan
- PI: [Prof. Hung-Yu Kao](#), NTHU, Taiwan

Teaching Assistant *Indian Institute of Technology Jammu*

*Jammu, India
Jan 2022 – March 2026*

- Assisted in six courses spanning core CS theory and systems: Machine Learning, Database Management Systems, Design and Analysis of Algorithms, Theory of Computation, Discrete Mathematical Structures, and Data Structures and Algorithms.
- Worked under Profs. [Suman Banerjee](#), [Shaifu Gupta](#), [Harkeerat Kaur](#), [Sumit Kumar Pandey](#).

Project Associate *Indian Institute of Technology (IIT) Bhilai*

*Chhattisgarh, India
Nov 2021 – Dec 2021*

- Project: *Building Trust on Computing Platform and Training of Secure Coding of Security Chips*, funded by MeitY, Government of India
- PI: [Prof. Dhiman Saha](#), IIT Bhilai, India

Graduate and Post-Graduate Students Guided

- [Devendra Mani](#), M.Tech Thesis (2027), IIT Jammu
- [Aditi Anand](#), B.Tech Project (2026), IIT Jammu
- [Ansh Jasrotia](#), Winter Intern (2026), NIT Srinagar
- [Abishek Salaria](#), Winter Intern (2026), NIT Srinagar
- [Paras Dhatwalia](#), Summer Intern (2025), Himachal Pradesh University
- [Yamini Uppal](#), AICTE-SAMARTHAN Intern (2025), SMVDU, Jammu
- [Amit Kumar](#), AICTE-SAMARTHAN Intern (2025), Bharati Vidyapeeth, Pune
- [Rajibul Islam](#), Rise-Up Intern (2025), GITA, Bhubaneswar
- [Harishchandra Kumar](#), Rise-Up Intern (2024), NIT Raipur
- [Tejash Gupta](#), Rise-Up Intern (2023), NIT Hamirpur
- [Ankit Kumar Bhagat](#), Rise-Up Intern (2022), University of Delhi
- [Atharva Sanjay Tekawade](#), Under-Graduate Research Program (2022), IIT Jammu

Professional Services

Journal Reviewer

- IEEE Transactions on Audio, Speech and Language Processing
- Knowledge and Information Systems (KAIS)
- GeoJournal
- Journal of Supercomputing

Conference Reviewer

- AAAI Conference on Artificial Intelligence (AAAI)

Technical Skills

Programming Languages: C, C++, Java, Python

Tools & Platforms: L^AT_EX, Linux, Microsoft SQL Server, Oracle Database, NetworkX, Gurobi/CPLEX

References

Prof. Suman Banerjee [✉](#)

Assistant Professor

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Prof. Shweta Jain [✉](#)

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Prof. Yamuna Prasad [✉](#)

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